Plant Image Classifier – End-to-End CNN Project

Deploy

LeafGuard

Choose page

Disease Detection

🌿 LeafGuard — Plant Disease Detector

Upload a leaf image

Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files

Sample_leaf.jpg 91.2KB

Uploaded Image

Prediction

Class: `Strawberry___Leaf_scorch`

Confidence

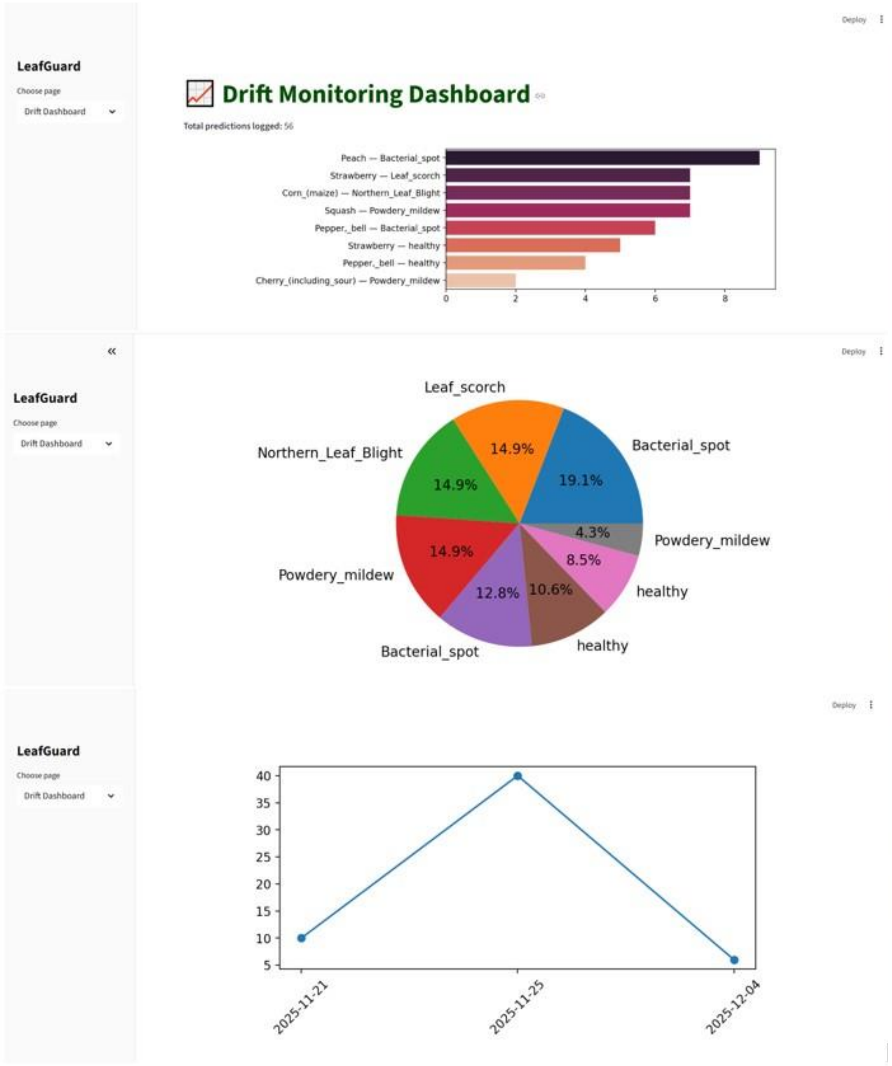
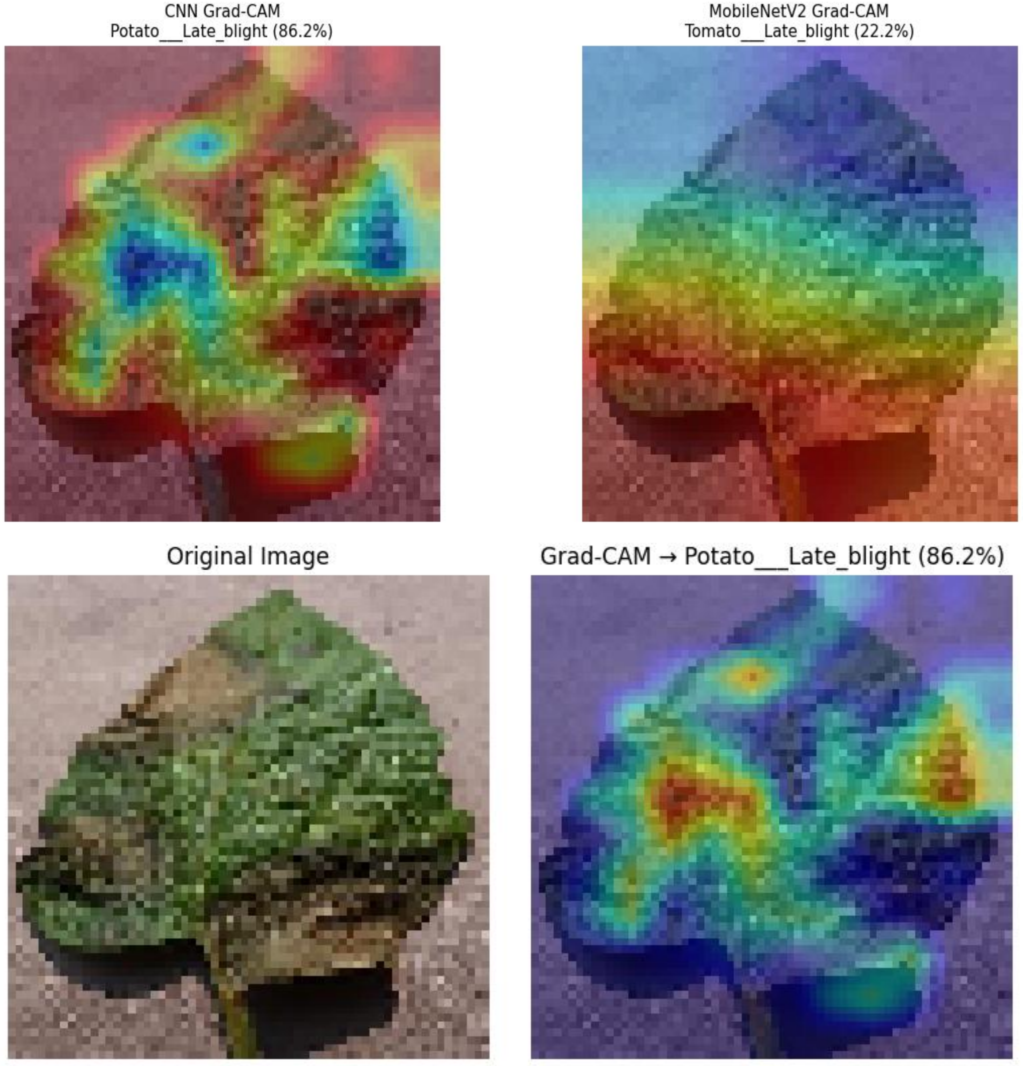
100.00%

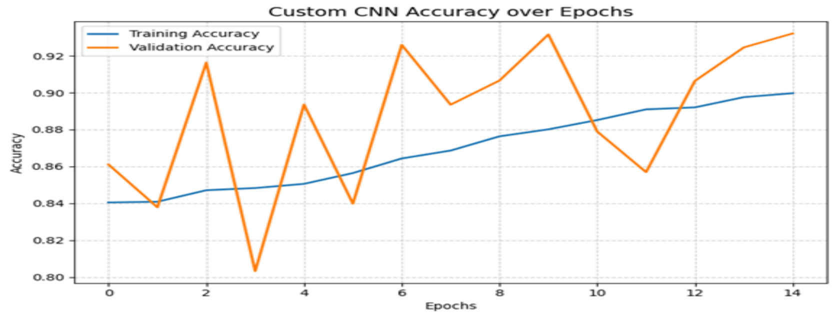
Status: Diseased

Disease Information

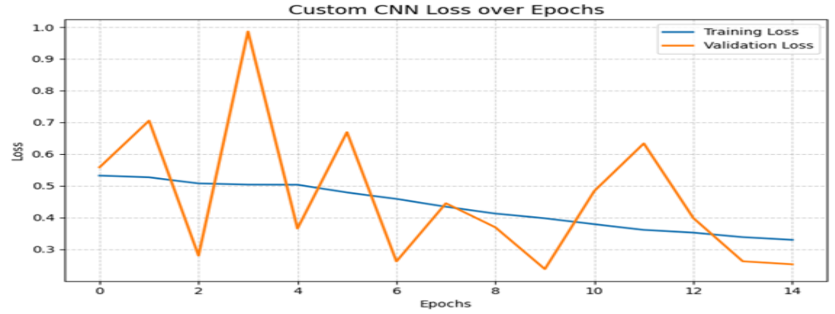
Description: Generated by AI

Cure / Treatment: To manage leaf scorch in strawberries, ensure proper watering by providing consistent moisture, especially during dry spells. Mulch around the plants to retain soil moisture and reduce temperature fluctuations. Additionally, avoid overhead watering to minimize leaf wetness and consider applying a balanced fertilizer to promote healthy growth.

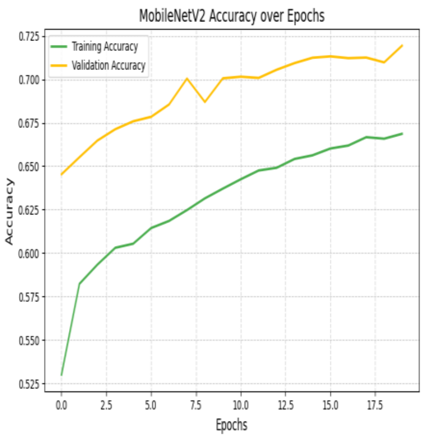

LeafGuard
Choose page
Drift Dashboard
Deploy
Drift Monitoring Dashboard
Total predictions logged: 56
Peach — Bacterial_spot
Strawberry — Leaf_scorch
Corn_(maize) — Northern_Leaf_Blight
Squash — Powdery_mildew
Pepper,_bell — Bacterial_spot
Strawberry — healthy
Pepper,_bell — healthy
Cherry_(including_sour) — Powdery_mildew
0
2
4
6
8
LeafGuard
Choose page
Drift Dashboard
Deploy
Leaf_scorch
Bacterial_spot
Northern_Leaf_Blight
Powdery_mildew
Powdery_mildew
healthy
Bacterial_spot
healthy
14.9%
19.1%
14.9%
4.3%
14.9%
8.5%
12.8%
10.6%
LeafGuard
Choose page
Drift Dashboard
Deploy
40
35
30
25
20
15
10
5
2025-11-21
2025-11-25
2025-12-04

CNN Grad-CAM
Potato___Late_blight (86.2%)
MobileNetV2 Grad-CAM
Tomato___Late_blight (22.2%)
Original Image
Grad-CAM → Potato___Late_blight (86.2%)



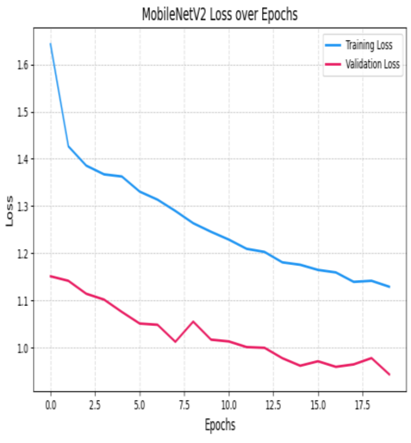
Custom CNN Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs



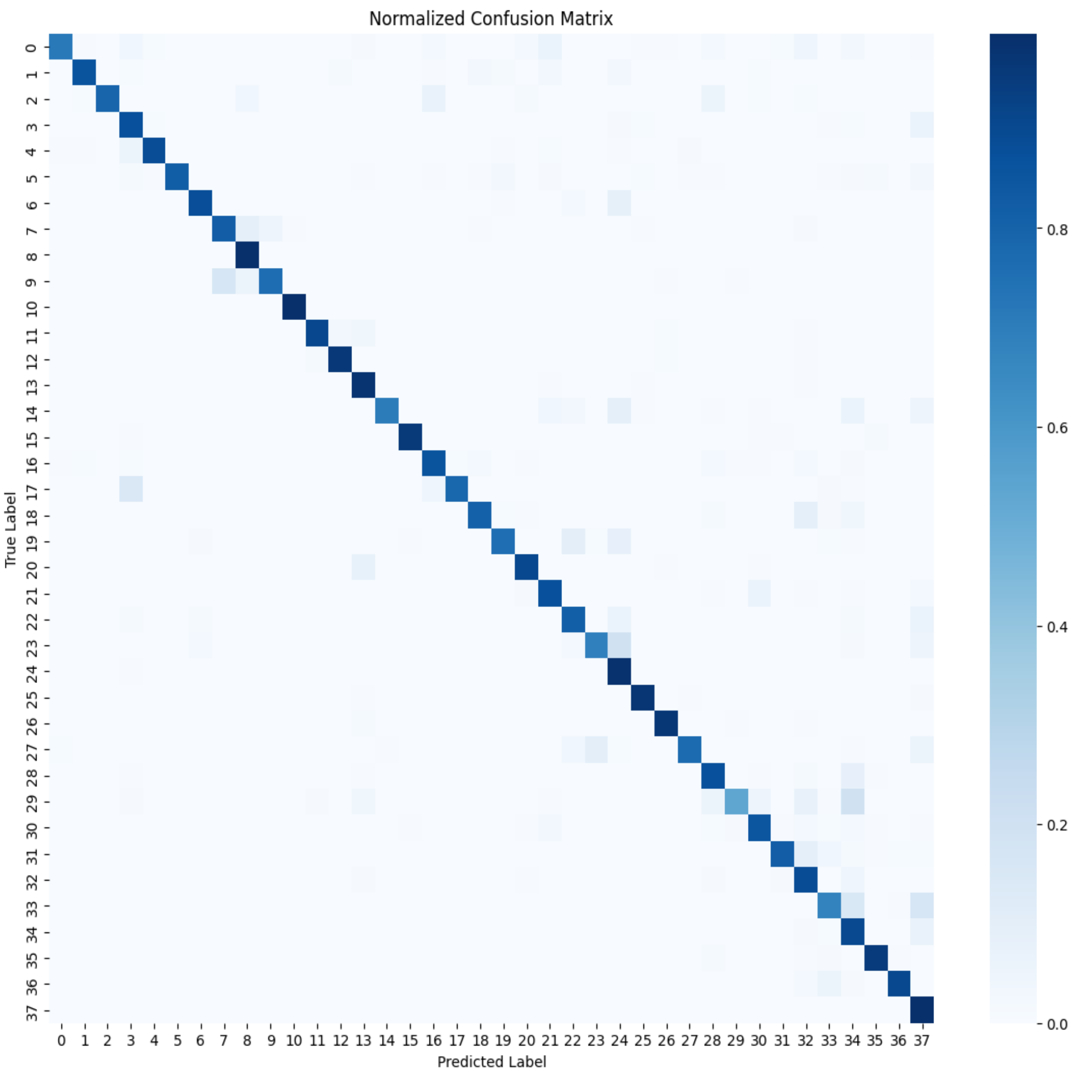
Custom CNN Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs



MobileNetV2 Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs



MobileNetV2 Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs



Normalized Confusion Matrix
True Label
Predicted Label

LeafGuard

Choose page

About

About LeafGuard

LeafGuard is an intelligent plant disease detection system built using a **Custom Convolutional Neural Network (CNN)** trained on the **PlantVillage dataset**.

🔬 What LeafGuard Does

- Identifies **38+ plant diseases** across fruits, vegetables, and crops
- Uses a trained **TensorFlow CNN model** for image classification
- Provides **AI-generated cures** using an LLM (OpenAI API, if available)
- Tracks prediction patterns using a **Drift Monitoring Dashboard**
- Detects changes in data trends using **distribution comparison & drift score**

🧠 Technologies Used

- **TensorFlow / Keras** (Custom CNN Image Classifier)
- **PlantVillage Dataset** (Training)
- **Streamlit** (Frontend UI + Dashboard)
- **Matplotlib & Seaborn** (Charts & visualizations)
- **OpenAI API** (Optional — to generate tailored disease treatments)

📁 App File Structure

- `models/best_custom_cnn_model.keras` — trained leaf disease model
- `drift_data/train_distribution.json` — class distribution from training
- `drift_data/drift_history.json` — logs all predictions
- `style.css` — custom UI styling

Project by **Iqra, Sameen & Laiba** 🌿